

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)		Light enters along a normal [to curved face] or equivalent		1		1		1
	(b)		35° 0.574 66° 0.914 (1) sin θ_p plotted between 0.57 and 0.58; sin θ_A between 0.91 and 0.92 (1) ecf		2		2	2	2
	(c)	(i)	Reasonable best fit line (straight) drawn. Accept straight line stated or as sin θ_p doubles, sin θ_A also doubles (1) Don't accept positive gradient Goes through origin (1) Stated to be little scatter of points about straight line (1) [so law confirmed]			3	3	1	3
		(ii)	Figures from a point on line put into $\frac{\sin \theta_A}{\sin \theta_p}$ (i.e. find gradient) (1) Above mark not available if n determined from a single point not on the line of best fit. n between 1.57 and 1.65 (1) (freestanding)			2	2	2	2
	(d)		40° is above critical angle or light would not refract out at the plane surface or tir would occur (1) Quantitative back-up, e.g. using n value from (c)(ii): $n \sin 40^\circ > 1$ or calculation of C : $n = 1.57, C = 39.5^\circ$; up to $n = 1.65, C = 37.3^\circ$, or taking inverse sine of horizontal scale reading where graph line hits top of grid or any other valid check ecf on n (1)		2		2	1	2
			Question 7 total	0	5	5	10	6	10